Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, angle θ has an opposite side of 5 and hypotenuse 13. Find $\sin\theta$ and the length of the adjacent side.

Adjacent $= \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$. $\sin\theta = \frac{5}{13}$.

2 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, angle θ has an opposite side of 15 and hypotenuse 17. Find $\cos\theta$.

Adjacent $= \sqrt{17^2 - 15^2} = \sqrt{289 - 225} = \sqrt{64} = 8$. $\cos\theta = \frac{8}{17}$.

3 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, angle θ has an adjacent side of 7 and hypotenuse 25. Find $\sin\theta$ and $\tan\theta$.

Opposite $= \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$. $\sin\theta = \frac{24}{25}$, $\tan\theta = \frac{24}{7}$.

4 **[Foundational]** • **Target: Right-triangle trigonometric ratios**

In a right triangle, the two legs are 20 and 21, and the hypotenuse is 29. Find $\sin\theta$ and $\cos\theta$ for the angle θ opposite the side of length 20.

$\sin\theta = \frac{20}{29}$, $\cos\theta = \frac{21}{29}$.

5 **[Foundational]** • **Target: Special angles**

State the exact value of $\sin(30^\circ)$.

$\sin(30^\circ) = \frac{1}{2}$.

6 **[Foundational]** • **Target: Special angles**

State the exact value of $\cos(45^\circ)$.

$\cos(45^\circ) = \frac{\sqrt{2}}{2}$.

7 **[Foundational]** • **Target: Special angles**

State the exact value of $\tan(60^\circ)$.

$\tan(60^\circ) = \sqrt{3}$.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Special angles

State the exact values of $\sin(90^\circ)$ and $\cos(0^\circ)$.

$\sin(90^\circ) = 1$ and $\cos(0^\circ) = 1$.

9 [Intermediate] • Target: Converting between radians and degrees

Convert 180° to radians.

$180^\circ = \pi$ radians.

10 [Intermediate] • Target: Converting between radians and degrees

Convert 90° to radians.

$90^\circ = \frac{\pi}{2}$ radians.

11 [Intermediate] • Target: Converting between radians and degrees

Convert $\frac{5\pi}{6}$ radians to degrees.

$\frac{5\pi}{6} \times \frac{180^\circ}{\pi} = 150^\circ$.

12 [Intermediate] • Target: Converting between radians and degrees

Convert 270° to radians.

$270^\circ = \frac{3\pi}{2}$ radians.

13 [Intermediate] • Target: The Pythagorean identity

Given $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is acute, $\cos\theta = \frac{4}{5}$.

14 [Intermediate] • Target: The Pythagorean identity

Given $\cos\theta = \frac{5}{13}$ with θ acute, find $\sin\theta$.

$\sin^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$. Since θ is acute, $\sin\theta = \frac{12}{13}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Pythagorean identity • Cross-Lesson: 7.01 Integration & Applications

Given $\sin\theta = 0.6$, find $\cos^2\theta$.

$\cos^2\theta = 1 - 0.36 = 0.64$.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Simplifying trigonometric expressions • Cross-Lesson: 8.01 Exponentials & Logarithms

Simplify $\sin^2\theta + \cos^2\theta + 1$.

$\sin^2\theta + \cos^2\theta = 1$, so the expression equals $1 + 1 = 2$.

17 [Challenging] • Target: Simplifying trigonometric expressions • Cross-Lesson: 9.01 Statistics & Probability

Simplify $\frac{1-\cos^2\theta}{\sin\theta}$ (for $\sin\theta \neq 0$).

$1 - \cos^2\theta = \sin^2\theta$, so the expression equals $\frac{\sin^2\theta}{\sin\theta} = \sin\theta$.

18 [Challenging] • Target: Simplifying trigonometric expressions • Cross-Lesson: 1.01 Algebra & Equations

Simplify $\tan\theta\cos\theta$ (for $\cos\theta \neq 0$).

$\tan\theta = \frac{\sin\theta}{\cos\theta}$, so $\tan\theta\cos\theta = \sin\theta$.

19 [Challenging] • Target: Graphs: amplitude and period • Cross-Lesson: 2.01 Functions & Transformations

State the amplitude and period of $y = 5\sin(x)$.

Amplitude: 5. Period: 2π.

20 [Challenging] • Target: Graphs: amplitude and period • Cross-Lesson: 3.01 Sequences & Series

State the amplitude and period of $y = 3\cos(2x)$.

Amplitude: 3. Period: $\frac{2\pi}{2} = \pi$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In the right triangle shown, angle θ has opposite side 8 and hypotenuse 17. Find $\sin\theta$.

a. $\frac{17}{8}$

b. $\frac{8}{15}$

c. $\frac{15}{17}$

d. $\frac{8}{17}$ ✓

$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{8}{17}$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In the right triangle shown, the two legs are 9 and 40, and the hypotenuse is 41. Find $\sin\theta$ and $\cos\theta$ (where θ is opposite the side of length 9).

a. $\sin\theta = \frac{9}{40}$, $\cos\theta = \frac{40}{9}$

b. $\sin\theta = \frac{40}{41}$, $\cos\theta = \frac{9}{41}$

c. $\sin\theta = \frac{41}{9}$, $\cos\theta = \frac{41}{40}$

d. $\sin\theta = \frac{9}{41}$, $\cos\theta = \frac{40}{41}$ ✓

Opposite $= 9$, adjacent $= 40$, hypotenuse $= 41$.

23 [Foundational] • Target: MCQ — Graphs: amplitude and period

The graph shows $y = A\sin(x)$ for some A. Find A by reading the graph's amplitude.

a. $A = 1$

b. $A = 2\pi$

c. $A = 4$

d. $A = 2$ ✓

The curve's peak height (amplitude) is 2.

24 [Foundational] • Target: MCQ — Graphs: phase shift and vertical shift

The graph shows $y = \sin(x)$ shifted vertically. Find the new midline by reading the graph.

a. $y = 0$

b. $y = 2$ ✓

c. $y = 1$

d. $y = -2$

The curve oscillates evenly about $y = 2$.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: MCQ — The unit circle: coordinates of key points

The unit circle shown marks a point at angle $\theta = 60^\circ$. What are its coordinates?

a. $\left(\frac{\sqrt{3}}{2}, 0.5\right)$

b. $(0.5, 0.5)$

c. $(1, 0)$

d. $\left(0.5, \frac{\sqrt{3}}{2}\right)$ ✓

$\cos(60^\circ) = 0.5$, $\sin(60^\circ) = \frac{\sqrt{3}}{2}$.

26 [Foundational] • Target: MCQ — Word problems: angles of elevation

A ladder leans against a wall, making an angle of 70° with the ground. If the ladder shown is 12 meters long, find how high up the wall it reaches, to the nearest meter.

a. 4

b. 12

c. 8

d. 11 ✓

$h = 12\sin(70^\circ) \approx 11$ m.

27 [Foundational] • Target: MCQ — Right-triangle trigonometric ratios

In a right triangle, angle θ has opposite side 9 and hypotenuse 41. Find $\cos\theta$.

a. $\frac{9}{41}$

b. $\frac{9}{40}$

c. $\frac{41}{40}$

d. $\frac{40}{41}$ ✓

Adjacent $= \sqrt{41^2 - 9^2} = 40$; $\cos\theta = \frac{40}{41}$.

28 [Foundational] • Target: MCQ — Special angles

State the exact value of $\sin(45^\circ)$.

a. 1

b. $\frac{\sqrt{3}}{2}$

c. $\frac{\sqrt{2}}{2}$ ✓

d. $\frac{1}{2}$

$\sin(45^\circ) = \frac{\sqrt{2}}{2}$.

© ClipSAT — clipsat.org | All Rights Reserved

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: MCQ — Converting between radians and degrees**

Convert 360° to radians.

a. 4π

b. π

c. 2π ✓

d. $\frac{\pi}{2}$

$360 \times \frac{\pi}{180} = 2\pi$.

30 **[Intermediate]** • **Target: MCQ — The Pythagorean identity**

Given $\sin\theta = \frac{4}{5}$ with θ acute, find $\cos\theta$.

a. $\frac{3}{5}$ ✓

b. $\frac{4}{5}$

c. $\frac{5}{3}$

d. $\frac{9}{25}$

$\cos^2\theta = 1 - \frac{16}{25} = \frac{9}{25} \Rightarrow \cos\theta = \frac{3}{5}$.

31 **[Intermediate]** • **Target: MCQ — Simplifying trigonometric expressions**

Simplify $\sin^2\theta + \cos^2\theta - 1$.

a. $2\sin^2\theta$

b. 0 ✓

c. 2

d. 1

By the Pythagorean identity, $\sin^2\theta + \cos^2\theta = 1$, so this is 0.

32 **[Intermediate]** • **Target: MCQ — Graphs: amplitude and period**

State the amplitude and period of $y = 2\cos(3x)$.

a. **Amplitude 2, period $\frac{2\pi}{3}$** ✓

b. Amplitude 3, period $\frac{2\pi}{2}$

c. Amplitude 6, period 2π

d. Amplitude 2, period 6π

Amplitude $= 2$; period $= \frac{2\pi}{3}$.

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: MCQ — Graphs: phase shift and vertical shift

State the vertical shift and midline of $y = 2\sin(x) - 4$.

a. Down 4, midline $y = 2$

b. Up 4, midline $y = 4$

c. **Down 4, midline** $y = -4$ ✓

d. Down 2, midline $y = -4$

Subtracting 4 shifts down 4; the midline is $y = -4$.

34 [Intermediate] • Target: MCQ — Solving basic trigonometric equations

Solve $2\cos\theta = 1$ for $0^\circ \le \theta < 360^\circ$.

a. $\theta = 30^\circ$ or $\theta = 330^\circ$

b. $\theta = 60^\circ$ or $\theta = 240^\circ$

c. $\theta = 60^\circ$ only

d. $\theta = 60^\circ$ **or** $\theta = 300^\circ$ ✓

$\cos\theta = 0.5$: $\theta = 60^\circ$ or 300°.

35 [Intermediate] • Target: MCQ — Reference angles in other quadrants

Find $\cos(210^\circ)$ using the reference angle.

a. $-\frac{\sqrt{3}}{2}$ ✓

b. $\frac{\sqrt{3}}{2}$

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

Reference angle 30°; cosine is negative in Q3: $\cos(210^\circ) = -\cos(30^\circ) = -\frac{\sqrt{3}}{2}$.

36 [Intermediate] • Target: MCQ — Complementary angle relationships

Given $\cos(40^\circ) \approx 0.766$, find $\sin(50^\circ)$.

a. 0.643

b. 0.766 ✓

c. 1.306

d. 0.234

$\sin(50^\circ) = \sin(90^\circ - 40^\circ) = \cos(40^\circ) \approx 0.766$.

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: MCQ — Word problems: angles of elevation

A drone hovers directly above a landmark. From a point 150 meters from the landmark's base, the angle of elevation to the drone is 40°. Find the drone's height, to the nearest meter.

a. 150

b. 196

c. 126 ✓

d. 115

$h = 150\tan(40^\circ) \approx 126$ m.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Graphs: amplitude and period • Cross-Lesson: 1.01 Algebra & Equations

State the exact value of $\sin(60^\circ)$.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{\sqrt{3}}{3}$

c. 1

d. $\sqrt{3}$

$\sin(60^\circ) = \frac{\sqrt{3}}{2}$.

39 [Challenging] • Target: MCQ — Graphs: phase shift and vertical shift • Cross-Lesson: 2.01 Functions & Transformations

In a right triangle, angle θ has opposite side 15, adjacent side 8, and hypotenuse 17. Find $\tan\theta$.

a. $\frac{3}{2}$

b. $\frac{8}{15}$

c. $\frac{15}{8}$ ✓

d. $\frac{13}{8}$

$\tan\theta = \frac{15}{8}$.

40 [Challenging] • Target: MCQ — Solving basic trigonometric equations • Cross-Lesson: 3.01 Sequences & Series

In a right triangle, angle θ has opposite side 48, adjacent side 14, and hypotenuse 50. Find $\tan\theta$.

a. 3

b. $\frac{24}{7}$ ✓

c. $\frac{23}{7}$

d. $\frac{7}{24}$

$\tan\theta = \frac{48}{14} = \frac{24}{7}$.

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: MCQ — Reference angles in other quadrants • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Given $\sin\theta = \frac{5}{13}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{1}{13}$

b. $\frac{14}{13}$

c. $\frac{10}{13}$

d. $\frac{12}{13}$ ✓

$\cos^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$. Since θ is acute, $\cos\theta = \frac{12}{13}$.

42 [Challenging] • Target: MCQ — Complementary angle relationships • Cross-Lesson: 6.01 Limits & Derivatives

State the amplitude and period of $y = 4\sin(4x)$.

a. Amplitude: 4. Period: $\frac{\pi}{2}$ ✓

b. Amplitude: 4. Period: $\frac{2\pi}{5}$

c. Amplitude: 5. Period: $\frac{2\pi}{5}$

d. Amplitude: 5. Period: $\frac{\pi}{2}$

Amplitude: 4. Period: $\frac{2\pi}{4} = \frac{\pi}{2}$.

43 [Challenging] • Target: MCQ — The unit circle: coordinates of key points • Cross-Lesson: 7.01 Integration & Applications

Solve $2\sin\theta = -1$ for $0^\circ \le \theta < 360^\circ$.

a. $\theta = 210^\circ$ **or** $\theta = 330^\circ$ ✓

b. $\theta = 210^\circ$ or $\theta = 60^\circ$

c. $\theta = 150^\circ$ or $\theta = 30^\circ$

d. $\theta = 300^\circ$ or $\theta = 330^\circ$

$\sin\theta = -\frac{1}{2}$. Solutions: $\theta = 210^\circ$ or $\theta = 330^\circ$.

44 [Challenging] • Target: MCQ — Word problems: angles of elevation • Cross-Lesson: 8.01 Exponentials & Logarithms

Find $\sin(210^\circ)$ using the reference angle.

a. $-\frac{\sqrt{3}}{3}$

b. $\frac{1}{2}$

c. $-\frac{1}{2}$ ✓

d. $-\frac{\sqrt{3}}{2}$

The reference angle is 30°; in Quadrant III, sine is negative. $\sin(210^\circ) = -\sin(30^\circ) = -\frac{1}{2}$.

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Statistics & Probability

Given $\cos\theta = \frac{7}{25}$ with θ acute, find $\sin\theta$.

a. $\frac{576}{625}$

b. $\frac{24}{25}$ ✓

c. $\frac{25}{24}$

d. $\frac{7}{25}$

$\sin^2\theta = 1 - \frac{49}{625} = \frac{576}{625} \Rightarrow \sin\theta = \frac{24}{25}$.

© ClipSAT — clipsat.org | All Rights Reserved